16. Given $f(x) = 5x - 2b$ while $g(x) = 4bx$. If $f\big(g(1)\big) = 36$, what is $g\big(f(1)\big)$?

2.7A Composition of Functions (Part 1)

AP Precalculus Name: _____ **CA #1**

Let $f(x) = 3^x$ and $g(x) = x^2 - x$.

1. Find $f\big(g(2)\big)$.

2. Find $g\big(f(-1)\big)$.

3. Find $(g \circ f)\left(\frac{1}{2}\right)$.

Let $f(x) = \frac{6}{x}$ and $g(x) = \sqrt{1-x}$.

4. Find $(f \circ g)(-3)$.

5. Find $g\big(f(-2)\big)$.

6. Find $g\big(f(6)\big)$.

7. Let $f(x) = 3 - x^2$ and $h(x) = f\big(g(x)\big)$. Fill in the table.

x	$g(x)$	$h(x)$
-1	2	
0	-1	
1	4	

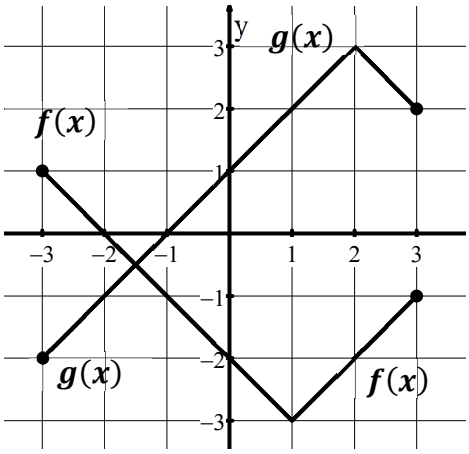
8. Let $g(x) = \frac{8}{x}$ and $h(x) = g\big(f(x)\big)$. Fill in the table.

x	$f(x)$	$h(x)$
2	2	
8	-4	
16	3	

9. Fill in the following table, given that $h(x) = f\big(g(x)\big)$.

x	$f(x)$	$g(x)$	$h(x)$
-7	2	6	
-3	6	-7	
-1	-3	-3	
0	5	2	
2	-7	5	
5	-1	0	
6	0	-1	

10. Use the graph to find each value.

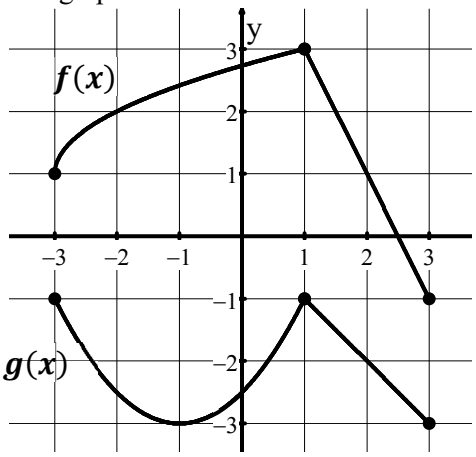


a. Find $f\big(g(2)\big)$.

b. Find $g \circ f(3)$.

c. Find $g\big(f(2)\big)$.

11. Use the graph to find each value.



a. Find $f\big(g(2)\big)$.

b. Find $f \circ g(-1)$.

c. Find $g\big(f(1)\big)$.

Answers to 2.7A CA #1

1. 9	2. $-\frac{2}{9}$	3. $3-\sqrt{3}$	4. 3	5. 2	6. 0

7a. -1 7b. 2 7c. -13	8a. 4 8b. -2 8c. $\frac{8}{3}$	9. $0, 2, 6, -7, -1, 5, -3$	10a. -1 10b. 0 10c. -1	11a. 2 11b. 1 11c. -3

Write your questions and thoughts here!

2.7B Composition of Functions (Part 2) — 2.7B Notes

Finding a Composed Function

1. Let $f(x) = x^2 - 1$ and $g(x) = \sqrt{x}$. Find the function $h(x) = f\big(g(x)\big)$.

2. Let $f(x) = 2x - x^2$ and $g(x) = x + 1$. Find the function $h(x) = f\big(g(x)\big)$.

Domain of a Composed Function

To find the domain of a composed function $f\big(g(x)\big)$, we have two restrictions to consider.

1. First, since we are using $g(x)$ as an input, we can only use x-values that are in the domain of g.
2. Second, the output of $g(x)$ must be in the domain of $f(x)$. To check this, we look at the new function $f\big(g(x)\big)$. Any restrictions on the domain of $f\big(g(x)\big)$ must also be included.

Use problem #1 and find the domain of $h(x)$.

3. Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 - 8$.
   a. Find $f \circ g$, then state the domain.

   b. Find $g\big(f(x)\big)$, then state the domain.

Write your questions and thoughts here!

Decomposition of Functions

Functions given analytically can often be decomposed into less complex functions.

Decompose the following functions as $h(x) = f(g(x))$.

4. $h(x) = \sqrt{x^3 + 1}$

$f(x) =$

$g(x) =$

5. $h(x) = \frac{1}{x^2+1}$

$f(x) =$

$g(x) =$

2.7B Composition of Functions (Part 2)

AP Precalculus

2.7B Practice

Let $f(x) = x + 2$ and $g(x) = \frac{1}{x}$.

1. Find $f \circ g$

2. State the domain of $f \circ g$

3. Find $g \circ f$

4. State the domain of $g \circ f$

Let $f(x) = \sqrt{4 - x}$ and $g(x) = x^2$.

5. Find $f \circ g$

6. State the domain of $f \circ g$

Let $f(x) = \sqrt{4 - x}$ and $g(x) = x^2$.

7. Find $g \circ f$

8. State the domain of $g \circ f$

Let $f(x) = \frac{3}{x^2}$ and $g(x) = \frac{1}{x}$.

9. Find $f \circ g$

10. State the domain of $f \circ g$

11. Find $g \circ f$

12. State the domain of $g \circ f$

Express h as a composition of two simpler functions f and g where $h(x) = f(g(x))$.

13. $h(x) = (2x - 7)^4$

$f(x) =$

$g(x) =$

14. $h(x) = \frac{4}{\sqrt{x}+3}$

$f(x) =$

$g(x) =$

15. $h(x) = e^{x^3}$

$f(x) =$

$g(x) =$

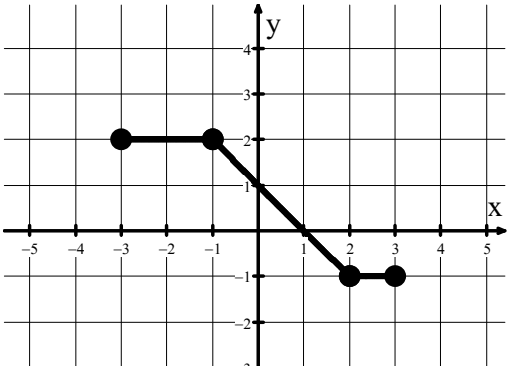
16. $h(x) = \sqrt{x^2 - x}$

$f(x) =$

$g(x) =$

17. Given that $f(x) = cx - 3$ and $g(x) = cx + 5$ are both defined on the set of all real numbers and c is a constant, what is the value of c if $(f \circ g)(x) = (g \circ f)(x)$ for all values of x?

18. The piecewise-linear function f, defined on $-3 \leq x \leq 3$, is shown in the graph. The function g is given by $g(x) = x - 2$. Sketch a graph of $y = f(g(x))$.



Graph of f

2.7B Composition of Functions (Part 2)

AP Precalculus Name: _____ **CA #1**

Let $f(x) = 2x$ and $g(x) = 8 - x^3$.

1. Find $f \circ g$

2. State the domain of $f \circ g$

3. Find $g \circ f$

4. State the domain of $g \circ f$

Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2$.

5. Find $f \circ g$

6. State the domain of $f \circ g$

7. Find $g \circ f$

8. State the domain of $g \circ f$

Express h as a composition of two simpler functions f and g where $h(x) = f(g(x))$.

9. $h(x) = e^{3x-1}$

$f(x) =$

$g(x) =$

10. $h(x) = \frac{1}{x^2-2x}$

$f(x) =$

$g(x) =$

Answers to 2.7B CA #1

1. $f \circ g = 16 - 2x^3$	2. All real numbers	3. $g \circ f = 8 - 8x^3$	4. All real numbers	5. $f \circ g = \sqrt{x^2 - 1}$
6. $(-\infty, -1][1, \infty)$	7. $g \circ f = x - 1$	8. $[1, \infty)$	9. $f(x) = e^x$ $g(x) = 3x - 1$	10. $f(x) = \frac{1}{x}$ $g(x) = x^2 - 2x$

AP Precalc

Write your questions and thoughts here!

2.8 Inverse Functions

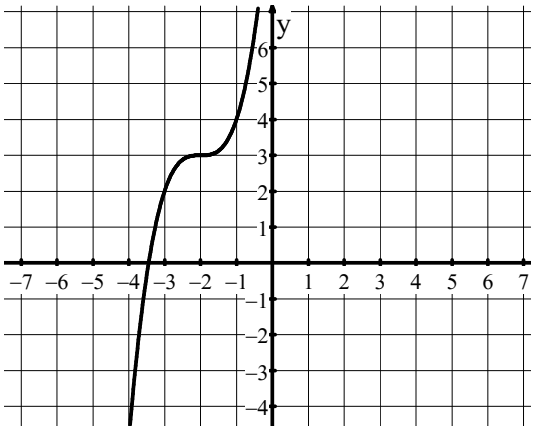
2.8 Notes

A function, f, has an inverse function, or is _____, if each output value of f is mapped from a unique input value.

$$f(x) = (x+2)^3 + 3$$

x	$f(x)$

x	$f^{-1}(x)$

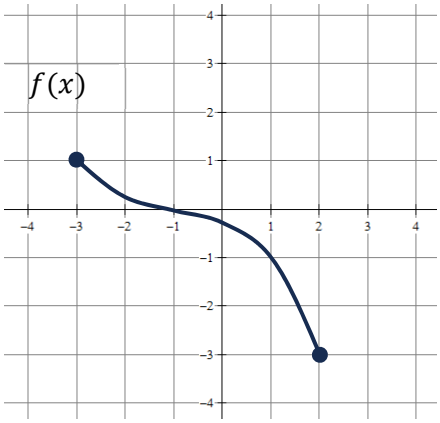


An inverse function is a reverse mapping of the function. That is, if $f(a) = b$, then $f^{-1}(b) = a$. Another way of thinking of this is if a function has the coordinate pair (a, b), then the inverse function has the coordinate pair (b, a)

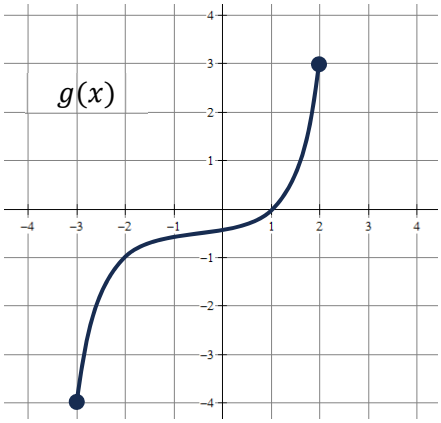
If the function is increasing or decreasing only, then it is invertible. If the graph "turns" (has a max or min), then it is no longer invertible because there will be output values that are the same for different input values. Think of this as a "horizontal line test". The **Vertical Line Test** checks to see if a graph is a function. The **Horizontal Line Test** checks to see if a graph's inverse is a function.

Are the following functions invertible? Sketch the graph of the inverse.

1.



2.

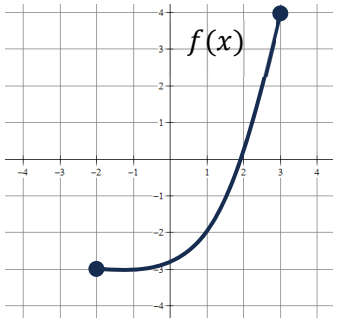


The inverse of the graph of the function $f(x)$ can be found by reversing the roles of the x- and y-axes. This means we can reflect the graph of f over the line $y = x$ to get the graph of the inverse.

The domain and range of a function and its inverse are swapped.

What is the minimum value of $f^{-1}(x)$?

What is the maximum value of $f^{-1}(x)$?



Write your questions and thoughts here!

One method of finding the inverse function is to reverse the roles of x and y in the equation, then solve for y.

3. Find the inverse function of $f(x) = (x+2)^3 + 3$.

The domain of a function can be restricted to make the function invertible.

4. Find the inverse function of $f(x) = \frac{1}{2}x^2 + 2$

What is the domain and range of the inverse function?

Find the inverse function along with the domain and range of the inverse.

5. $f(x) = \sqrt{x-3} + 2$	6. $f(x) = \frac{3}{x+6}$	7. $f(x) = \frac{2x-3}{x+4}$
Domain of f^{-1}.	Domain of f^{-1}.	Domain of f^{-1}.
Range of f^{-1}.	Range of f^{-1}.	Range of f^{-1}.

Composition of f and f^{-1}

The composition of a function, f, and its inverse function f^{-1}, is the identity function.

$$f\left(f^{-1}(x)\right) =$$

8. Are $f(x) = \frac{2}{x+3}$ and $g(x) = \frac{2}{x} - 3$ inverses?

2.8 Inverse Functions

AP Precalculus

2.8 Practice

Find the inverse of each function and list the domain and range of $f^{-1}(x)$.

1. $f(x) = (x-3)^3 + 4$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

2. $f(x) = \frac{1}{7}x + 6$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

3. $f(x) = (x+1)^2 - 2$ for $x \geq -1$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

4. $f(x) = \sqrt{x+2} - 3$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

5. $f(x) = (x-2)^2 + 5$ for $x \leq 2$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

6. $f(x) = \frac{2}{x-1}$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

7. $f(x) = -(x+4)^2 - 1$ for $x \leq -4$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

8. $f(x) = -\sqrt{x+1} + 3$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

9. $f(x) = \frac{2x+5}{3x-4}$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

Use the tables below to find the given values.

10.

x	$f(x)$
1	-2
2	3
3	6
4	2
5	4
6	1

a. $f(1)$
b. $f(6)$
c. $f^{-1}(1)$
d. $f^{-1}(4)$
e. $f(2)$
f. $f^{-1}(6)$

11.

x	$f(x)$
-3	2
-1	7
2	10
7	-3
8	-1
10	8

a. $f(2)$
b. $f(10)$
c. $f^{-1}(7)$
d. $f^{-1}(-3)$
e. $f(7)$
f. $f^{-1}(2)$

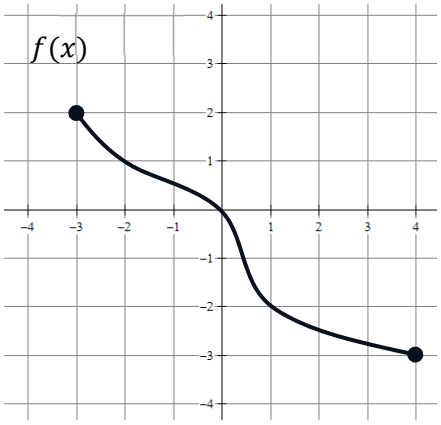
12.

x	$f(x)$
-10	-6
-6	7
-2	-10
3	11
7	3
11	-2

a. $f(-10)$
b. $f(3)$
c. $f^{-1}(7)$
d. $f^{-1}(-6)$
e. $f(7)$
f. $f^{-1}(-2)$

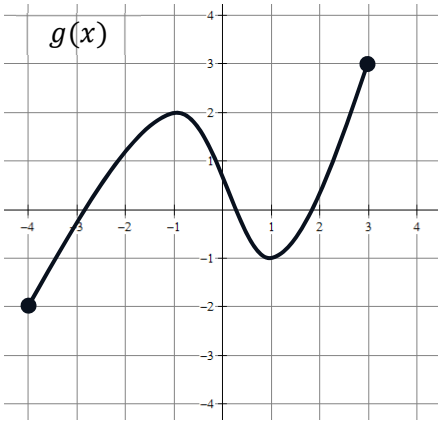
The graph of a function is given below. Identify if the function is invertible. Sketch the graph of the inverse regardless of whether or not it is invertible.

13.



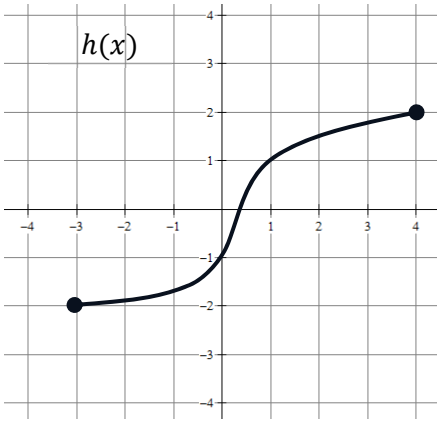
Is $f(x)$ invertible?

14.



Is $g(x)$ invertible?

15.



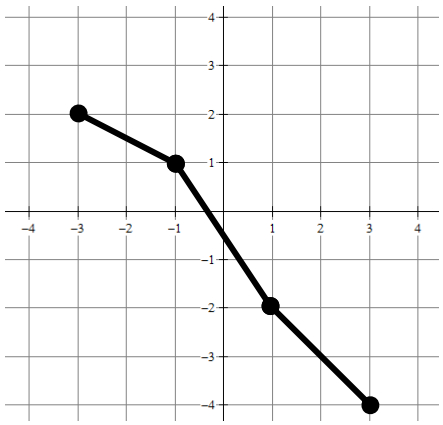
Is $h(x)$ invertible?

Determine if the two functions are inverses of each other using composition.

16. $f(x) = 3x + 5$ and $g(x) = \frac{1}{3}x - \frac{5}{3}$

17. $f(x) = \sqrt[3]{3 - x}$ and $g(x) = x^3 - 3$

18. The graph of the piecewise-linear function f is shown in the figure. Let g be the inverse function of f. What is the minimum value of g?



Graph of f

(A) -4

(B) -3

(C) 2

(D) 3

---

19. Mr. Brust is filling up his backdoor kiddie pool with the water hose. The amount of water, in gallons, in the pool t minutes after he turns on the water can be modeled by P, an increasing function of time t. Which of the following gives a verbal representation of the function P^{-1}, the inverse of P?

(A) P^{-1} is an increasing function of the amount of time after the water is turned on.

(B) P^{-1} is a decreasing function of the amount of time after the water is turned on.

(C) P^{-1} is an increasing function of the amount of water in the pool.

(D) P^{-1} is a decreasing function of the amount of water in the pool.

20.

x	1	2	3	4	5
$f(x)$	-18	-10	-3	1	26

Calculator active. Let f be an increasing function for $x \geq 0$. The table gives values of $f(x)$ at selected values of x. The function g is given by $g(x) = \frac{x^4+16x^3+50}{x-2}$

a. The function h is defined by $h(x) = (g \circ f)(x) = g\big(f(x)\big)$. Find the value of $h(2)$ as a decimal approximation or indicate that it is not defined.

b. Find the value of $f^{-1}(1)$, or indicate that it is not defined.

2.8 Inverse Functions

AP Precalculus Name: _____ **CA #1**

Find the inverse of each function and list the domain and range of $f^{-1}(x)$.

1. $f(x) = (x-8)^2 + 4$ for $x \geq 8$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

2. $f(x) = (x+9)^2 - 6$ for $x \leq -9$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

3. $f(x) = \sqrt{x-7} + 1$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

4. $f(x) = -\sqrt{x-1} + 3$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

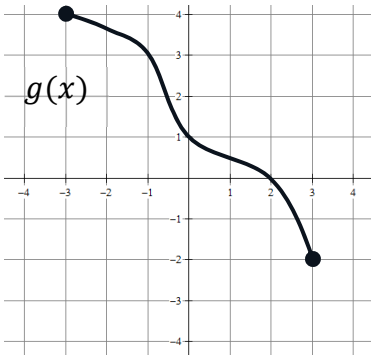
5. $f(x) = \frac{2}{x+5}$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

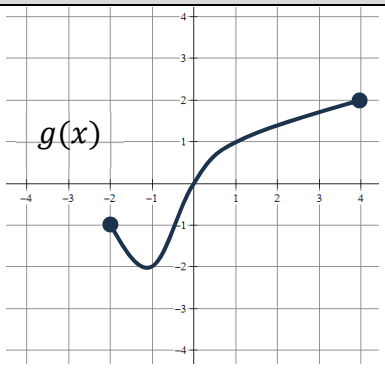
The graph of a function is given below. Identify if the function is invertible. Sketch the graph of the inverse regardless of whether or not it is invertible.

6.



Is $g(x)$ invertible?

7.



Is $g(x)$ invertible?

Use the tables below to find the given values.

8.

x	$f(x)$
1	6
2	4
3	2
4	1
5	3
6	5

a. $f(1)$

b. $f(6)$

c. $f^{-1}(1)$

d. $f^{-1}(4)$

e. $f(2)$

f. $f^{-1}(6)$

9.

x	$f(x)$
-7	-2
-2	13
0	1
1	6
6	-7
13	0

a. $f(1)$

b. $f(13)$

c. $f^{-1}(-2)$

d. $f^{-1}(0)$

e. $f(-7)$

f. $f^{-1}(6)$

Answers to 2.8 CA #1

1. $f^{-1}(x) = \sqrt{x-4}+8$
Domain: $x \geq 4$
Range: $y \geq 8$

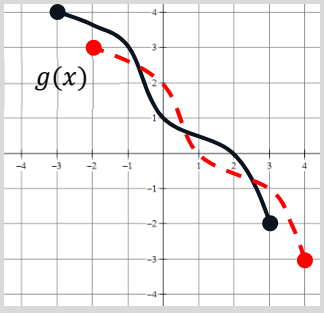
2. $f^{-1}(x) = -\sqrt{x+6}+9$
Domain: $x \geq -6$
Range: $y \leq -9$

3. $f^{-1}(x) = (x-1)^2+7$
Domain: $x \geq 1$
Range: $y \geq 7$

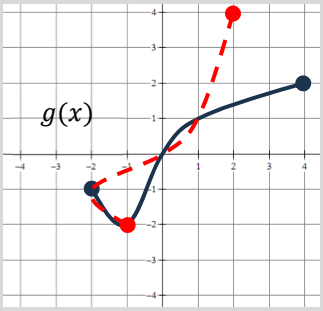
4. $f^{-1}(x) = (x-3)^2+1$
Domain: $x \leq 3$
Range: $y \geq 1$

5. $f^{-1}(x) = \frac{2}{x} - 5$
Domain: $\mathbb{R}$, $x \neq 0$
Range: $\mathbb{R}$, $y \neq -5$

6. Yes, it is invertible



7. No, it is not invertible.



8a. 6
8b. 5
8c. 4
8d. 2
8e. 4
8f. 1

9a. 6
9b. 0
9c. -7
9d. 13
9e. -2
9f. 1

Name: _____ Date:_____ Period: _____ **2A Review**

Unit 2-A Review – Exponential and Logarithmic Functions

Reviews do NOT cover all material from the lessons but will hopefully remind you of key points. To be prepared, you must study all packets for lessons 2.1 – 2.8.

Find an equation that gives the *n*th term of each sequence. Instead of the initial value use the *k*th term of the sequence in your equation. *k* is given for each problem.

1. $\{116, 136, 156, 176, \ldots\} \quad k = 2$

2. $\left\{2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \ldots\right\} \quad k = 1$

3. For the sequence $\{8, 5, 2, -1, \ldots\}$, what is the 315th term?

A function has the following coordinate points. Could the function represent a linear function, exponential function, or neither?

4. $(3, 6), (4,11), (5,16)$

5. $(-10, 3), (-9, 6), (-8, 12)$

6. The following function is either linear or exponential. Which is it? Justify your answer.

x	6	11	16
$f(x)$	10	13	16

Is each function linear or exponential. Identify the constant (slope or ratio) that causes the output values to change?

7. $y = -\left(\frac{3}{5}\right)^x$

8. $y - 7 = 6(x - 4)$

9. It is known that y is an exponential function and that it passes through the points $(3, 6)$ and $(9, 42)$. Write an equation for this function.

10. The value $4 \cdot 4 \cdot 4 \cdot 6$ is an output value of an exponential function of the form $f(x) = a \cdot b^x$, where a and b are constants. Write the function along with the input value that represents the output value.

$f(x) =$

where $x =$

11. Identify if the exponential function is growth or decay and justify your response. $f(x) = 1.4(0.07)^x$

12. Is the exponential function $f(x) = 8(1.7)^x$ increasing or decreasing?

13. Is the exponential function $f(x) = -(6)^x$ concave up or concave down?

14. Find $\lim_{x \to -\infty} f(x)$ for the function $f(x) = 4(0.3)^x$

Let $f(x)$ be a function on which a transformation occurs. Let $g(x)$ be a transformation of f. For each problem, name the transformation(s) of f.

15. $f(x) = 9^x$ and $g(x) = f(x) \cdot 9$

16. $f(x) = 2^x$ and $g(x) = \frac{f(x)}{16} \cdot$

17. $f(x) = 7^x$ and $g(x) = \big(f(x)\big)^{-3}$

18. $f(x) = 7^x$ and $g(x) = -\frac{f(x)}{49}$

Evaluate the function at the given input values.

19. Let $h(x) = 6 \cdot 4^{x/3}$. Find $h(2)$

20. Let $h(x) = 9 \cdot 2^{x/5}$. Find $h(-1)$

21. Below is a table of values for an exponential function in the form $f(x) = a(b)^x + k$. Write the equation for this exponential function.

x	0	1	2	3	4
$f(x)$	3	6	18	66	258

22. The table gives the weight of a catfish at selected weeks t during its first eight weeks of life.

Week	1	3	6	8	9
Weight (pounds)	0.08	0.2	0.467	0.501	0.7

a. Use an exponential regression $f(t) = ab^t$ to model these data. Round to three decimals but store the original equation in your calculator.

b. According to the model in your calculator, how much will the catfish weigh after 5 weeks?

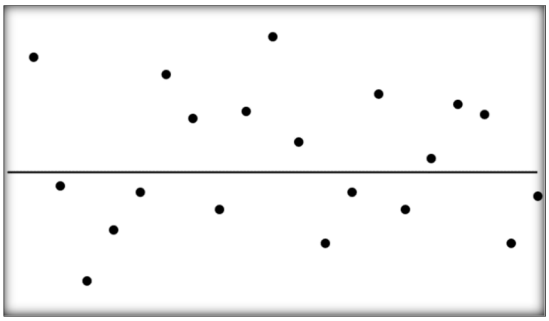
c. According to the model in your calculator, how much will the catfish weigh after 7 weeks?

d. If the model continues past 9 weeks, when will the catfish weigh 1 pound?

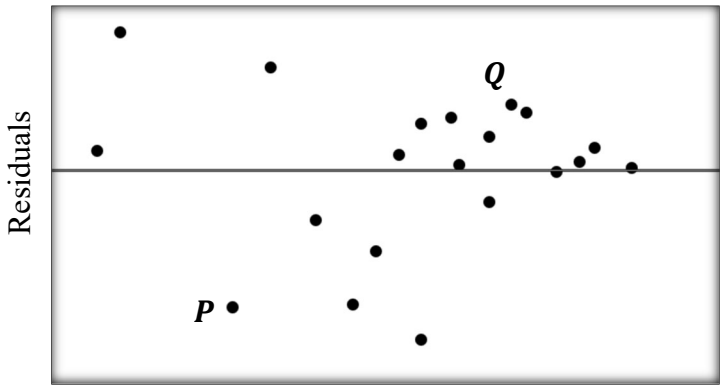
e. Use your model from part a to find the error at $t = 8$.

f. Is the value predicted an overestimate or underestimate of the actual value?

23. A student creates a model for a data set. The residual plot for their exponential regression is shown. Is the model appropriate? Why or why not?



24. Several students' SAT scores are collected and compared to their GPA from their first year at college. A regression model was calculated with the SAT scores as input values and GPA from the first year at college as the output values. The residual plot for the regression model and data points is below.



a. The given residual plot has a point labeled P at the coordinate $(1050, -0.633)$. What does point P indicate in the context?

b. The point labeled Q is at the coordinate $(1420, 0.3)$. What does point Q indicate in the context?

25. Use the tables to find the given values.

x	$g(x)$
-5	4
-2	10
0	-2
4	8

x	$f(x)$
-5	5
-2	9
0	0
4	-2

a. $g(f(4)) =$

b. $(f \circ g)(0) =$

Let $f(x) = 3x^2 - 4x$ and $g(x) = 5 - 3x$.

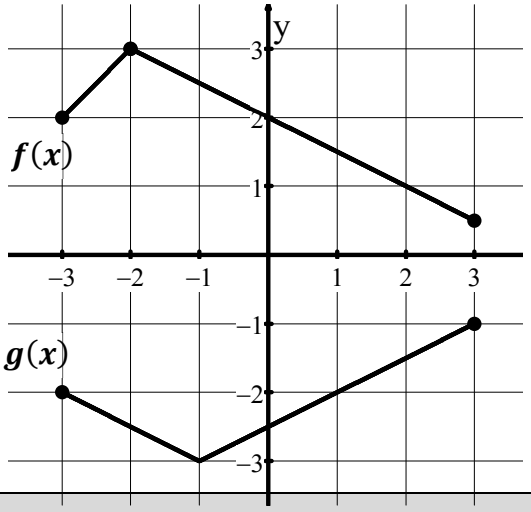
26. Find $f\big(g(-1)\big)$.

27. Find $(g \circ f)(1)$.

28. Let $f(x) = 7x - 5$ and $h(x) = f\big(g(x)\big)$. Fill in the table.

x	$g(x)$	$h(x)$
-5	2	
0	-1	
5	3	

Use the graph to find the following values.



29. Find $f\big(g(1)\big)$.

30. Find $f\big(g(-3)\big)$.

31. Find $g\big(f(2)\big)$.

Let $f(x) = \frac{1}{x-6}$ and $g(x) = 1 - x$.

32. Find $f \circ g$

33. State the domain of $f \circ g$

34. Find $g \circ f$

35. State the domain of $g \circ f$

36. Express h as a composition of two simpler functions f and g where $h(x) = f\big(g(x)\big)$.

$$h(x) = \sqrt{7 - x^2}$$

$f(x) =$

$g(x) =$

37. Find the inverse of $f(x) = (x - 2)^2 + 1$ for $x \leq 2$. List the domain and range of $f^{-1}(x)$.

Name: _____ Date:_____

Corrective Assignment

Unit 2A CA – Exponential and Logarithmic Functions

Find an equation that gives the *n*th term of each sequence. Instead of the initial value use the *k*th term of the sequence in your equation. *k* is given for each problem.

1. $\{-2, -8, -32, -128, \ldots\}$ $k = 1$

2. $\{325, 225, 125, 25, \ldots\}$ $k = 4$

3. $\{-25, -19, -13, -7, \ldots\}$ $k = 0$

4. $\left\{16, 4, 1, \frac{1}{4}, \ldots\right\}$ $k = 3$

Find the *n*th term of each sequence. Write an equation for each sequence before finding the nth term.

5. $\{117, 106, 95, 84, \ldots\}$ What is the 47th term?

6. $\{-24, -19, -14, -9, \ldots\}$ What is the 401st term?

A function has the following coordinate points. Could the function represent a linear function, exponential function, or neither?

7. $(4, 6), (5, 10), (6, 18)$

8. $(1, 3), (2, 12), (3, 48)$

9. $(16, -5), (17, -1), (18, 3)$

The following functions are either linear or exponential. Which is it? Justify your answer.

10.

x	2	7	12
$f(x)$	3	12	48

11.

x	2	6	10
$f(x)$	12	5	-2

Is each function linear or exponential. Identify the constant (slope or ratio) that causes the output values to change?

12. $y = \left(\frac{2}{3}\right)^x$

13. $y = 9x - 1$

14. $y - 6 = \frac{1}{4} \cdot 2^{x-5}$

15. $y + 5 = 3(x + 6)$

It is known that $f(x)$ is a linear function and that it passes through the given points. Write an equation for this function.

16. $(8, 85)$ and $(13, 5)$

17. $(2, 8)$ and $(4, 20)$

It is known that $f(x)$ is an exponential function and that it passes through the given points. Write an equation for this function.

18. $(8, 85)$ and $(13, 5)$

19. $(2, 8)$ and $(4, 20)$

20. Identify if the exponential function $f(x) = 0.6(5.25)^x$ is growth or decay and justify your response.

21. The value $3 \cdot 3 \cdot 3 \cdot 3 \cdot 7$ is an output value of an exponential function of the form $f(x) = a \cdot b^x$, where a and b are constants. Write the function along with the input value that represents the output value.

$f(x) =$

where $x =$

22. Is the exponential function $f(x) = -4(0.6)^x$ increasing or decreasing?

23. Is the exponential function $f(x) = 105\left(\frac{2}{3}\right)^x$ concave up or concave down?

24. If $f(x) = -\frac{4}{5}(7)^x$, find…

a. $\lim_{x \to -\infty} f(x) =$

b. $\lim_{x \to \infty} f(x) =$

Let $f(x)$ be a function on which a transformation occurs. Let $g(x)$ be a transformation of f. For each problem, name the transformation(s) of f.

25. $f(x) = 2^x$ and $g(x) = f(x) \cdot 16$

26. $f(x) = 4^x$ and $g(x) = \big(f(x)\big)^{-2}$

27. $f(x) = 2^x$ and $g(x) = \frac{f(x)}{4}$.

28. $f(x) = 2^x$ and $g(x) = -\frac{f(x)}{16}$

Evaluate the function at the given input values.

29. Let $h(x) = 10 \cdot 5^{x/4}$. Find $h(2)$

30. Let $h(x) = 4 \cdot 8^{x/3}$. Find $h(-1)$

31. Below is a table of values for an exponential function in the form $f(x) = a(b)^x + k$. Write the equation for this exponential function.

x	0	1	2	3	4
$f(x)$	4	11	25	53	109

32. After a long weekend away, Mr. Brust comes home to find he left sliced bananas on the countertop, and fruit flies are all over his kitchen. Using his security cameras, he estimates the number of fruit flies in his kitchen at selected values of h, hours after Saturday morning at 8:00 a.m.

Hours (h)	1	10	15	24
Population of fruit flies	2	20	50	200

a. Use an exponential regression $F(h) = ab^h$ to model these data. Round to three decimals but store the original equation in your calculator.

b. According to the model in your calculator, how many fruit flies were there after 12 hours?

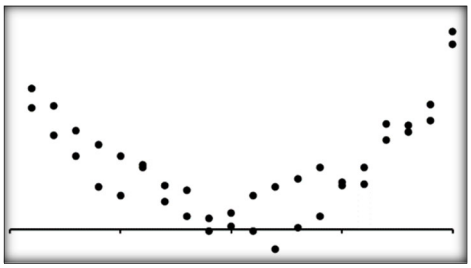
c. If the model were to continue, how many fruit flies would be in Mr. Brust's kitchen if he was gone for 3 days? Round to the nearest whole number.

d. How long before there are 1,000 fruit flies? Round to the nearest whole number.

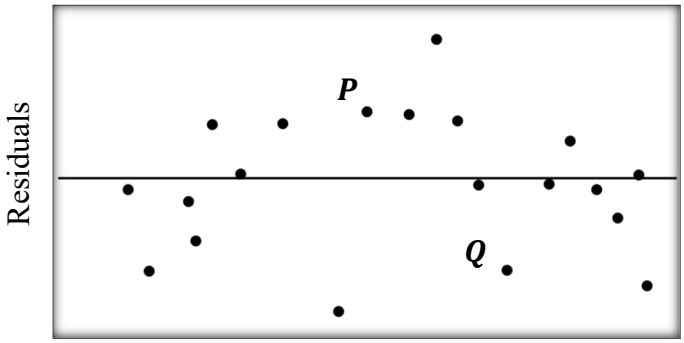
e. Use your model from part a to find the error at $t = 10$.

f. Is the value predicted an overestimate or underestimate of the actual value?

33. A student creates a model for a data set. The residual plot for their quadratic regression is shown. Is the model appropriate? Why or why not?



34. Mr. Sullivan loves going to the movies, and realizes that the more he makes, the more often he goes to the movie theater. He calculated a regression model with his yearly salary as input values and the amount he spends at a movie theater each year as the output values. The residual plot for the regression model and data points is below.



a. The given residual plot has a point labeled P at the coordinate $(67000, 37.09)$. What does point P indicate in the context?

b. The point labeled Q is at the coordinate $(77000, -51.91)$. What does point Q indicate in the context?

Let $f(x) = 4^x$ and $g(x) = -3 - x$.

35. Find $(f \circ g)(-2)$.

36. Find $g\left(f\left(\frac{1}{2}\right)\right)$.

37. Find $g(f(2))$.

38. Let $f(x) = 2^x + 1$ and $h(x) = f\big(g(x)\big)$. Fill in the table.

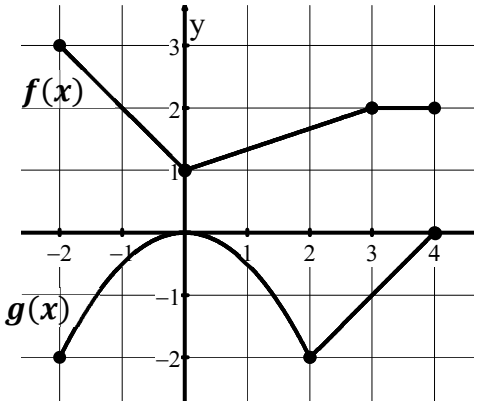
x	$g(x)$	$h(x)$
1	-1	
2	0	
3	2	

Use the graphs of f and g to find the given values.

39. Find $f\big(g(0)\big)$.

40. Find $g\big(f(3)\big)$.

41. Find $g\big(f(-2)\big)$.



Let $f(x) = \sqrt{2x-1}$ and $g(x) = 1 - x^2$.

42. Find $f \circ g$

43. State the domain of $f \circ g$

44. Find $g \circ f$

45. State the domain of $g \circ f$

46. Express $h(x) = e^{1-3x}$ as a composition of two simpler functions f and g where $h(x) = f\big(g(x)\big)$.

$f(x) =$

$g(x) =$

Find the inverse of each function. List the domain and range of $f^{-1}(x)$.

47. $f(x) = -\sqrt{x-2} + 7$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

48. $f(x) = \frac{6}{x-1}$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

49. $f(x) = (x+9)^2 + 6$ for $x \geq -9$

Domain of $f^{-1}(x)$:

Range of $f^{-1}(x)$:

50. Use the table to find the given values.

a. $f(6) =$

b. $f(11) =$

c. $f^{-1}(8) =$

d. $f^{-1}(6) =$

e. $f(9) =$

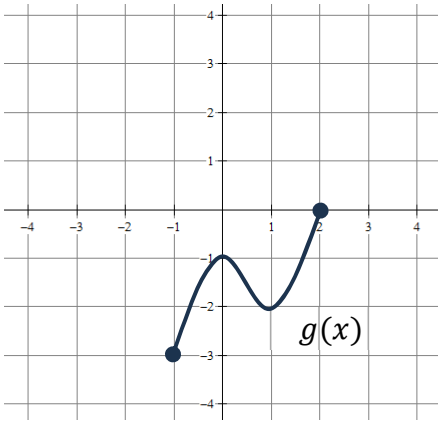
f. $f^{-1}(9) =$

x	$f(x)$
6	8
7	10
8	6
9	11
10	9
11	7

51.

a. Is $g(x)$ invertible?

b. Sketch the graph of $g^{-1}(x)$, regardless of whether or not it is invertible.



Answers to Unit 2A Corrective Assignment

1. $g_n = -2(4)^{n-1}$

2. $g_n = 25 - 100(n-4)$

3. $g_n = -31 + 6n$

4. $g_n = \left(\frac{1}{4}\right)^{n-3}$

5. -389

6. 1976

7. neither

8. exponential

9. linear

10. Exponential because for each input change of 5, f changes proportionally by a ratio of 4.

11. Linear because for each input change of 4, f changes at a constant rate -7.

12. Ratio of $\frac{2}{3}$

13. Slope of 9

14. Ratio of 2

15. Slope of 3

16. $y - 85 = -16(x-8)$ or $y - 5 = -16(x-13)$

17. $y - 8 = 6(x-2)$ or $y - 20 = 6(x-4)$

18. $y = 85 \cdot \left(\sqrt[5]{\frac{1}{17}}\right)^{x-8}$ or $y = 5 \cdot \left(\sqrt[5]{\frac{1}{17}}\right)^{x-13}$

19. $y = 8 \cdot \left(\sqrt{\frac{5}{2}}\right)^{x-2}$ or $y = 20 \cdot \left(\sqrt{\frac{5}{2}}\right)^{x-4}$

20. Growth because $b > 1$

21. $f(x) = 7(3)^x$ where $x = 4$

22. Increasing

23. Concave up

24.
a. $\lim_{x\to-\infty} f(x) = 0$
b. $\lim_{x\to\infty} f(x) = -\infty$

25. Shift left 4 units.

26. Horizontal dilation and reflection across the y-axis.

27. Shift right 2 units.

28. Reflection across the x-axis and shift right 4.

29. $10\sqrt[4]{25}$

30. $\frac{4}{\sqrt[3]{8}}$

31. $f(x) = 7(2)^x - 3$

32a. $F(h) = 2.078(1.221)^x$
32b. 22.762
32c. 3,586,274
32d. $x = 30.965$. 31 hours
32e. $15.304 - 20 = -4.696$
32f. Underestimate

33. No. The residual plot creates a pattern.

34a. Because point P is above the x-axis, for the year Mr. Sullivan's salary was \$67,000, the model underestimates the money spent at a movie theater by \$37.09.
34b. Because point Q is below the x-axis, for the year Mr. Sullivan's salary was \$77,000, the model overestimates the money spent at a movie theater by \$51.91.

35. $\frac{1}{4}$

36. -5

37. -19

38a. $\frac{3}{2}$
38b. 2
38c. 5

39. 1

40. -2

41. -1

42. $\sqrt{1-2x^2}$

43. $\left[-\sqrt{\frac{1}{2}}, \sqrt{\frac{1}{2}}\right]$

44. $2 - 2x$

45. $\left[\frac{1}{2}, \infty\right)$

46. $f(x) = e^x$
$g(x) = 1 - 3x$

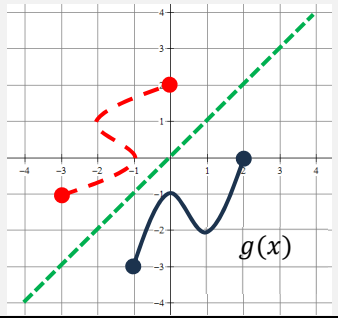
47. $f^{-1}(x) = (x-7)^2 + 2$
Domain: $x \le 7$
Range: $y \ge 2$

48. $f^{-1}(x) = \frac{6}{x} + 1$
Domain: $\mathbb{R}$, $x \neq 0$
Range: $\mathbb{R}$, $y \neq 1$

49. $f^{-1}(x) = \sqrt{x-6} - 9$
Domain: $x \ge 6$
Range: $y \ge -9$

50a. 8
50b. 7
50c. 6
50d. 8
50e. 11
50f. 10

51. Not invertible.



AP Precalc

2.9 Logarithic Expressions

2.9 Notes

Write your questions and thoughts here!

BIG IDEA

$$\log_b c = a$$
if and only if
$$b^a = c$$
$$b > 0 \text{ and } b \neq 1$$

Let's rewrite the following logarithms as exponents.

Ex 1: $\log_3 81 = 4$

Ex 2: $\log_{16} 4 = \frac{1}{2}$

Let's rewrite the following exponents as logarithms.

Ex 3: $125^{\frac{1}{3}} = 5$

Ex 4: $10^3 = 1000$

Find the value of each log.

Ex 5: $\log 100$

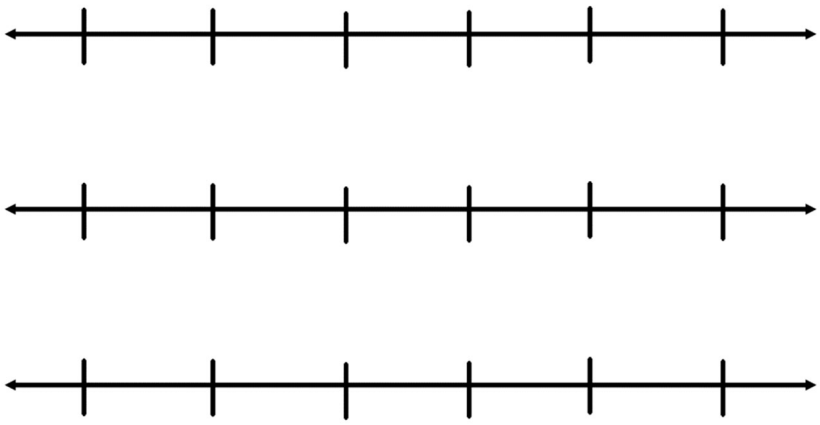
Ex 6: $\log_2 \frac{1}{32}$

Ex 7: $\log_4 25$

Ex 8: $\log 39$

Write your questions and thoughts here!

Algebro	Average Length of Video in seconds
Sully	505 seconds
Brust	830 seconds
Bean	1,440 seconds
Kelly	10,160 seconds



Try These:

Rewrite the exponent as a log.

1. $16^{\frac{3}{4}} = 8$

Rewrite the log as an exponent

2. $\log_{32} 4 = \frac{2}{5}$

Evaluate each log

3. $\log_{25} x$ when $x = 5$

4. $\log_{6} x$ when $x = 12$

2.9 Logarithmic Expressions

AP Precalculus

2.9 Practice

Rewrite the following logarithms as exponents.

1. $\log_2 64 = 6$

2. $\log_4 \frac{1}{64} = -3$

3. $\log_{25} 125 = \frac{3}{2}$

Rewrite the following exponents as logarithms.

4. $10^3 = 1000$

5. $16^{\frac{5}{2}} = 1024$

6. $10^{-2} = \frac{1}{100}$

WITHOUT using a CALCULATOR, find the value of logarithm. SHOW WORK.

7. $\log_5 625$	8. $\log 100{,}000$	9. $\log_{27} 81$

Use a CALCULATOR to find the value of logarithm. Round to three decimal places.

10. $\log 140$	11. $\log_9 120$	12. $\log_3 18$

For the given data construct a plot using a LOGARITHMIC scale using the given bases. Be sure to label your axis and show your math.

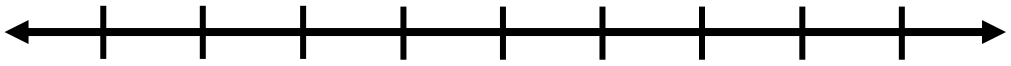
12. Logarithmic Scale of base 10.

Person	Midichlorians Found in Cells
Chewbacca	1
Han Solo	330
Luke Skywalker	125,000
Yoda	2,750,000
Anakin Skywalker	20,000



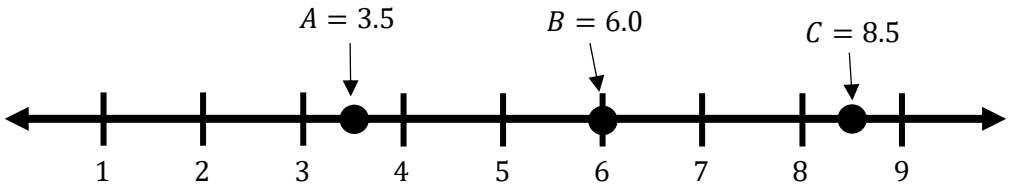
13. Logarithmic Scale of Base 2.

Person	Errors Per Section
Bean	5
Brust	68
Sullivan	24
Kelly	15
New Teacher	300



The Richter Scale is a common way of measuring earthquakes around the world. The scale measures the amplitude of the waves from seismic activity. The Richter Scale uses a logarithmic scale of base 10. This means each order of magnitude is 10 times greater than the previous one. A 6.0 earthquake is ten times more intense than a 5.0 earthquake.

Use the scale below to answer the questions that follow.



14. Which statement most accurately describes how much more intense Earthquake C was than Earthquake A.

(A) Earthquake C is 5 times more intense.

(B) Earthquake C is 50 times more intense.

(C) Earthquake C is 10^5 more intense.

(D) Earthquake C is $\log 5$ times more intense.

---

15. Which of the following statements is true.

(A) The increase in intensity from Earthquake A to B is greater than the increase in intensity from Earthquake B to C.

(B) The increase in intensity from Earthquake A to B is less than the increase in intensity from Earthquake B to C.

(C) Earthquake B is as many times more intense to Earthquake A, as Earthquake C is as many times more intense to Earthquake B.

2.9 Logarithmic Expressions

AP Precalculus Name: _____ **CA #1**

Directions: Rewrite the following logarithms as exponents.

1) $log_4 8 = \frac{3}{2}$

2) $\log \frac{1}{1000} = -3$

Directions: Rewrite the following exponents as logarithms.

3) $25^{\frac{3}{2}} = 125$

4) $6^3 = 216$

Directions: WITHOUT using a CALCULATOR, find the value of logarithm.

5) $log_3 \frac{1}{81}$

6) $log_2 128$

Directions: Use a CALCULATOR to find the value of logarithm. Round to three decimal places.

7) $log_8 6200$

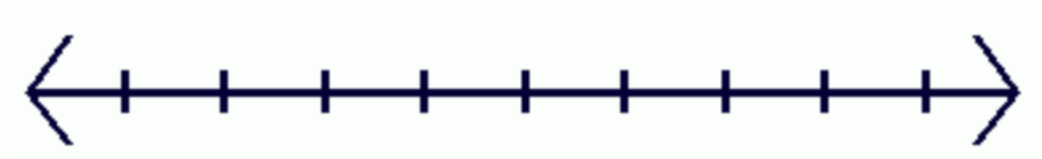
8) $\log 438$

Directions: For the given data construct a plot using a LOGARITHMIC scale using the given bases. Be sure to label your axis and show your math..

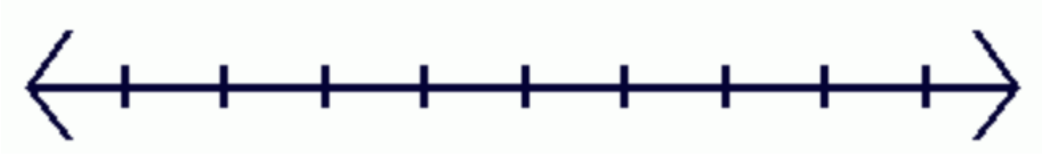
9)

Item	Weight in Milligrams
Pencil	5 mg
Eraser	45 mg
Calculator	6000 mg
Backpack	375,000 mg

Base 5



Base 10



Answers to 1.1 CA #1

1. $4^{\frac{3}{2}} = 8$	2. $10^{-3} = \frac{1}{1000}$	3. $log_{25}125 = \frac{3}{2}$	4. $log_6 216 = 3$	5. -4
6. 7	7. 4.199	8. 2.641	9. SEE BELOW	

9.

Item	Weight in Milligrams
Pencil	5 mg
Eraser	45 mg
Calculator	6000 mg
Backpack	375,000 mg

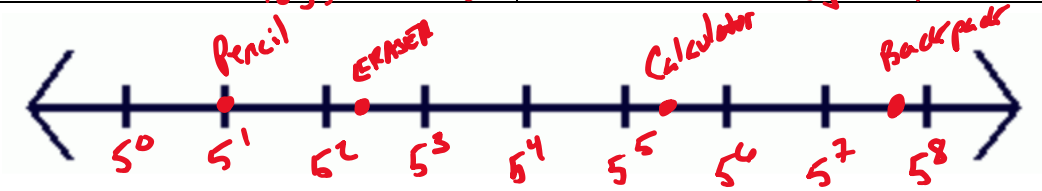
$\log_5 5 = 1$ $\log 5 = 0.699$

$\log_5 45 = 2.365$ $\log 45 = 1.653$

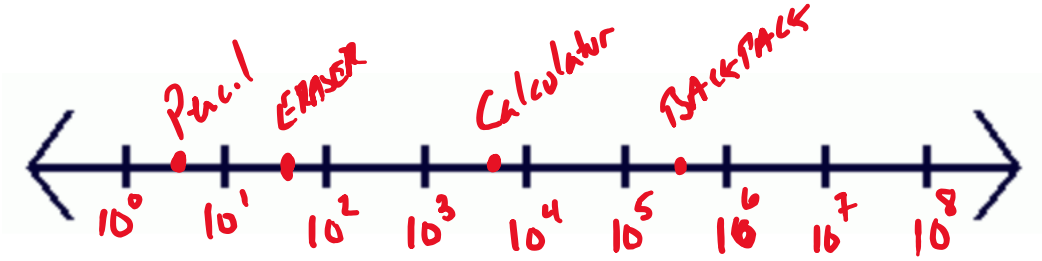
$\log_5 6000 = 5.405$ $\log 6000 = 3.778$

$\log_5 375{,}000 = 7.975$ $\log 375{,}000 = 5.574$

Base 5



Base 10



AP Precalc

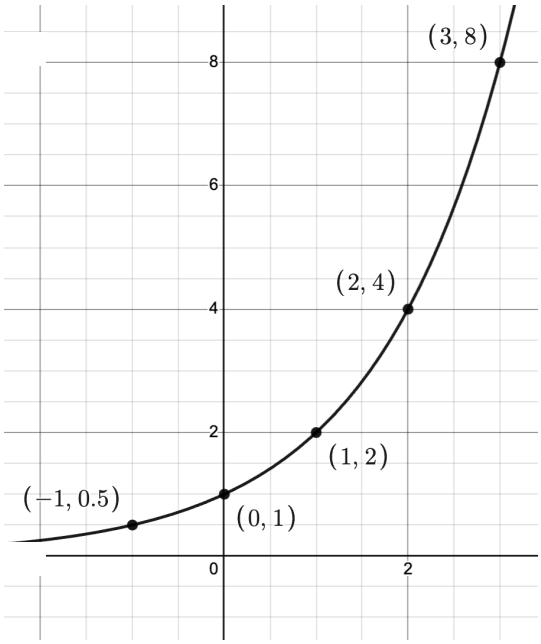
2.10 Inverses of Exponential Functions

2.10 Notes

Write your questions and thoughts here!

Exponential $f(x) = 2^x$

x	$f(x)$



Logarithmic $g(x) = \log_2 x$

x	$f(x)$

BIG IDEA

- The graphs are a reflection across the function
- The domain of the exponential function is the range of the logarithmic function.
- The range of the exponential function is the domain of the logarithmic function.
- The operations undo each other (like multiplication and division)
- $f(g(x)) = g(f(x)) = x$

Describe the function $f(x)$ as exponential or logarithmic and then find points for its inverse, $g(x)$.

Ex 1:

x	$f(x)$
-1	$\frac{1}{3}$
0	1
1	3
2	9
3	27

x	$g(x)$

Ex 2:

x	$f(x)$
$\frac{1}{4}$	-1
1	0
4	1
16	2
64	3

x	$g(x)$

Write your questions and thoughts here!

INVERSES UNDO EACH OTHER

$\mathbf{2^x}$ $\mathbf{\log_2 x}$

Are the following functions inverses of each other?

$f(x) = 2^x$ $g(x) = \log_2 x$

In other words, does $f\big(g(x)\big) = g\big(f(x)\big) = x$?

Find the inverse of each of the following.

Ex 3: $f(x) = 3\log_5 x$

Ex 4: $f(x) = 10^{\frac{x}{6}}$

Try these:

1. Determine if $f(x)$ and $g(x)$ are inverses.
   $f(x) = 5^{4x}$ and $g(x) = 4\log_5 x$
2. Find the inverse of $f(x) = 10\log_2 x$.

2.10 Inverses of Exponential Functions

AP Precalculus

2.10 Practice

Describe the function $f(x)$ as exponential, logarithmic, or neither and how you know why it is that function. Then find points for its inverse $g(x)$.

1.

x	$f(x)$
3	8
4	16
5	32
6	64

x	$g(x)$

2.

x	$f(x)$
$\frac{1}{9}$	-2
$\frac{1}{3}$	-1
1	0
3	1

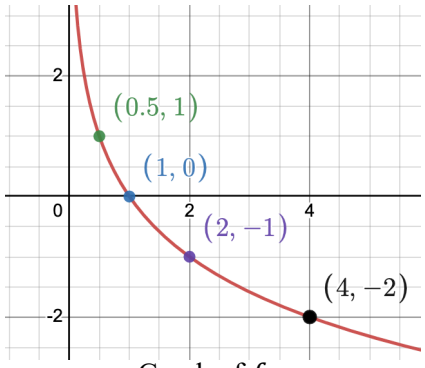
x	$g(x)$

3.

x	$f(x)$
0	1
1	4
2	7
3	10

x	$g(x)$

4.



Graph of f

x	$g(x)$

Determine if $f(x)$ and $g(x)$ are inverses.

5. $f(x) = 2\log_2 x$ and $g(x) = 2^{2x}$

6. $f(x) = 10^{0.25x}$ and $g(x) = 4\log x$

7. $f(x) = 5^{\frac{x}{8}}$ and $g(x) = 8\log_5 x$

Find the inverse of the given function.

8. $h(x) = 4^{5x}$

9. $m(x) = 5\log_2 x$

Find the inverse of the given function.

10. $a(x) = \frac{1}{4}\log_8 x$

11. $b(x) = 10^{\frac{3}{4}x}$

2.10 Inverses of Exponential Functions

2.10 Test Prep

12. Which of the following represents a possible function that is the inverse of $f(x) = 0.25^x$?

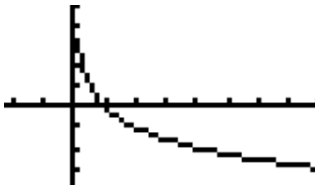
(A)

x	$f(x)$
-3	64
-2	16
-1	4
0	1

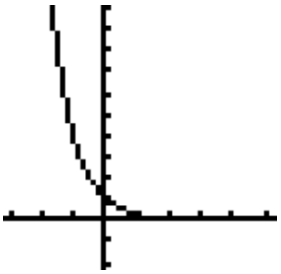
(B)

x	$f(x)$
4	-1
1	0
$\frac{1}{4}$	1
$\frac{1}{16}$	2

(C)



(D)



2.10 Inverses of Exponential Functions

AP Precalculus Name: _____ **CA #1**

Directions: Describe the function, f(x) (exponential, logarithmic, or neither), how you know why it is that function and then find points for its inverse, g(x).

1)

X	f(x)
8	3
16	4
32	5
64	6

X	G(x)

2)

X	f(x)
-3	1/8
-2	1/4
-1	1/2
0	1

X	G(x)

Directions: Determine if f(x) and g(x) are inverses.

3. $f(x) = 3 \cdot \log_5 x$
$g(x) = 5^{3x}$

4. $f(x) = 10^{5x}$
$g(x) = \frac{1}{5} \cdot \log x$

Directions: Find the inverse of the given function.

5. $h(x) = 3^{\frac{x}{10}}$

6. $m(x) = 6 \cdot log_5 x$

Answers to 1.1 CA #1

1. Logarithmic because the x-values are changing multiplicatively.

X	f(x)
3	8
4	16
5	32
6	64

2. Exponential because the y-values are changing multiplicatively.

X	f(x)
1/8	-3
¼	-2
½	-1
1	0

3. Not inverses

4. Inverses

5. $h^{-1}(x) = 10 \cdot log_3 x$

6. $m^{-1}(x) = 5^{\frac{x}{6}} = 5^{\frac{1}{6}x}$

AP Precalc

Write your questions and thoughts here!

2.11 Logarithmic Functions

2.11 Notes

Exponential Graph

x-intercept:

y-intercept:

asymptote:

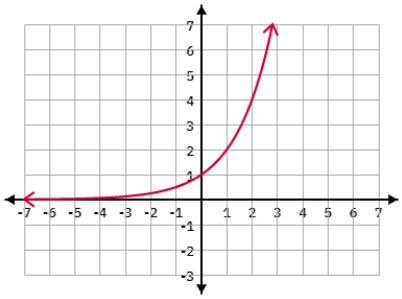
increasing:

decreasing:

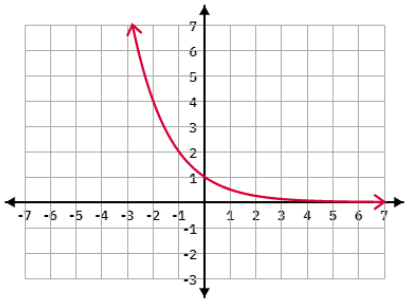
domain:

range:

$b > 1$



$0 < b < 1$



Logarithmic Graph

x-intercept:

y-intercept:

asymptote:

increasing:

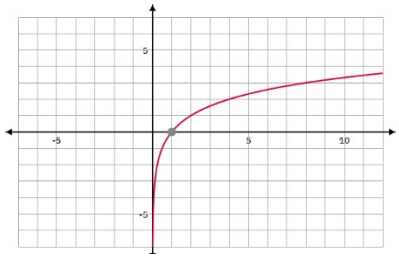
decreasing:

domain:

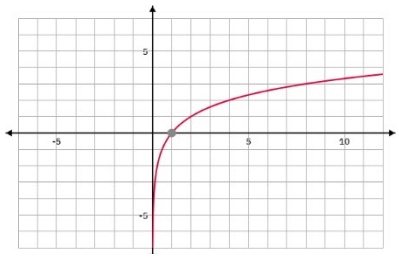
range:

Transformations of Logarithm Graphs

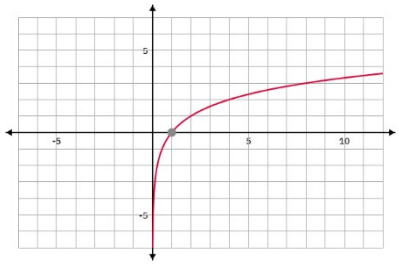
Horizontal/Vertical Translations



Horizontal/Vertical Dilations



Horizontal/Vertical Reflections



Which transformations affect the domain/range?

Write your questions and thoughts here!

Find the domain/range, asymptotes and end behavior of the following functions. (Sketch a graph to help)

a. $f(x) = \ln(4 - x) + 2$

b. $f(x) = -2\log_3(x - 2)$

USE A CALCULATOR

c. $f(x) = \log|x - 5|$

2.11 Logarithmic Functions

AP Precalculus

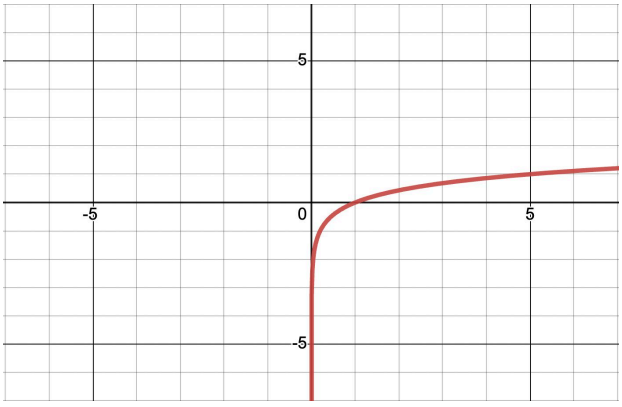
2.11 Practice

Sketch a graph of the transformation of $f(x) = \log_5 x$ onto the graph. Label each graph.

1. $g(x) = 3\log_5(x + 2) - 4$

2. $h(x) = 3\log_5(3 - x) + 1$

3. $j(x) = -\log_5(x - 3) - 2$



Find all relevant information from the given function. Sketch a graph. No calculator.

4. $f(x) = \ln(x - 3) + 5$

Asymptote:

Domain: Range:

End Behavior:

Graph:

5. $f(x) = -2\log_2(x + 3) - 6$

Asymptote:

Domain: Range:

End Behavior:

Graph:

6. $f(x) = \log_6(8 - x) + 1.5$

Asymptote:

Domain: Range:

End Behavior:

Graph:

7. $f(x) = -\log(2x - 5)$

Asymptote:

Domain: Range:

End Behavior:

Graph:

8. $f(x) = 455\log_9(x + 376) + 543$

Asymptote:

Domain: Range:

End Behavior:

Graph:

9. $f(x) = -3\ln(10 - x)$

Asymptote:

Domain: Range:

End Behavior:

Graph:

Write a logarithmic function with the given information.

10. End Behavior

$$\lim_{x\to\infty} f(x) = \infty$$
$$\lim_{x\to -6^+} f(x) = -\infty$$

11. Domain

$(-\infty, 8)$

CALCULATOR ACTIVE: Find all relevant information.

12. $f(x) = \log(|x + 4|) - 10$

Asymptote:

Domain: Range:

End Behavior:

13. $f(x) = -3\ln\left(\frac{x+7}{x}\right)$

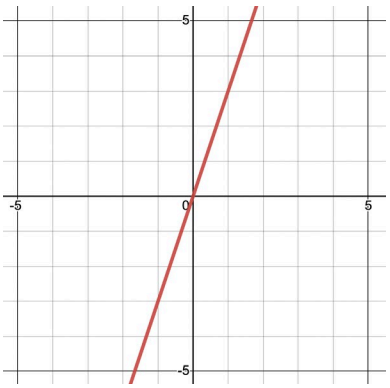
Asymptote:

Domain: Range:

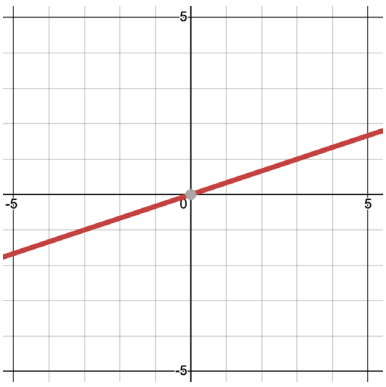
End Behavior:

14. The function h has the relationship that when the input is tripled the output values will increase by 1. Which of the following graphs could be $h(x)$?

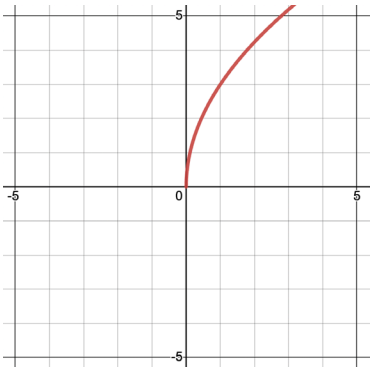
(A)



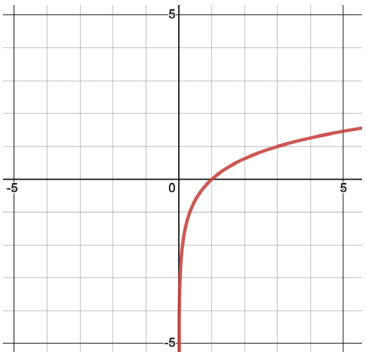
(B)



(C)



(D)



---

15. Which of the functions could have the following conditions?

$$\lim_{x\to-\infty} f(x) = -\infty$$
$$\lim_{x\to10^-} f(x) = \infty$$

(A) $f(x) = -3\ln(x-10)+5$

(B) $g(x) = -2\log(5-x)+10$

(C) $h(x) = -\log_3(10-x)-5$

(D) $j(x) = \log(10-x)+4$

---

16. **Calculator active.** If $\log_{0.2}(x+2) < \log_{0.04}(x+2)$, then x lies in which of the following intervals?

(A) $(-\infty,-1)$

(B) $(-2,-1)$

(C) $(-1,\infty)$

(D) $(1,2)$

2.11 Logarithmic Functions

AP Precalculus Name: _____ **CA #1**

Instructions: Find all relevant information from the given function. Sketch a graph. No calculator.

1) $f(x) = \ln(x-5) - 2$

Asymptote:

Domain: Range:

End Behavior:

Graph:

2) $f(x) = -5\ log_6(x+2) - 9$

Asymptote:

Domain: Range:

End Behavior:

Graph:

3) $f(x) = \log(7-x) + 4$

Asymptote:

Domain: Range:

End Behavior:

Graph:

4) $f(x) = -2\ log_2(2-x) + 4$

Asymptote:

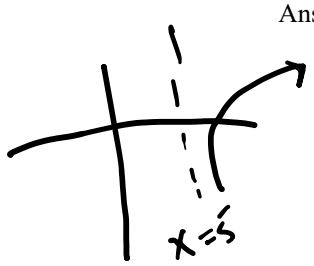
Domain: Range:

End Behavior:

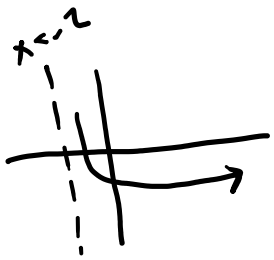
Graph:

Answers to 2.11 CA #1

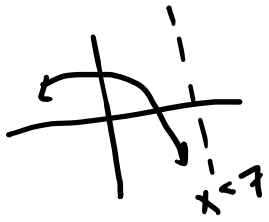
1) A: x = 5
D: $(5, \infty)$
R: $(-\infty, \infty)$
EB:
$\lim_{x\to\infty} f(x) = \infty$
$\lim_{x\to 5^+} f(x) = -\infty$



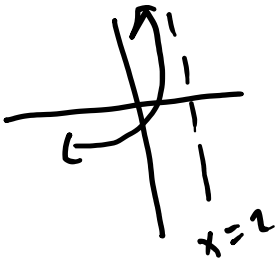
2) A: x = -2
D: $(-2, \infty)$
R: $(-\infty, \infty)$
EB:
$\lim_{x\to\infty} f(x) = -\infty$
$\lim_{x\to -2^+} f(x) = \infty$



3) A: x = 7
D: $(-\infty, 7)$
R: $(-\infty, \infty)$
EB:
$\lim_{x\to 7^-} f(x) = -\infty$
$\lim_{x\to -\infty} f(x) = \infty$



4) A: x = 2
D: $(-\infty, 2)$
R: $(-\infty, \infty)$
EB:
$\lim_{x\to 2^-} f(x) = \infty$
$\lim_{x\to -\infty} f(x) = -\infty$



AP Precalc

Write your questions and thoughts here!

2.12 Logarithmic Function Manipulation

2.12 Notes

Properties of Logarithmic Functions

	Property	But also…
Product Property:	$\log_b(xy) = \log_b x + \log_b y$	
Quotient Property:	$\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$	
Power Property:	$\log_b x^k = k\log_b x$	

Expanding and Condensing

a. $\log(4x^2y)$

b. $2\log_2 x + \frac{1}{2}\log_2(y+1)$

c. $\log\left(\frac{\sqrt[3]{x+5}}{y^2}\right)$

d. $2\log_3\left(ab^2\right) - 3\log_3 c + \frac{1}{3}\log_3 d$

Change of Base Formula

LOG

$$\log_b x = \frac{\log_a x}{\log_a b}$$

LN

$$\log_4 12$$

*Dirty little secret MATH